

Subject Name: Logistics and Supply Chain Management
Session No: 01
Faculty Name: Prof Ajit M Maurya

	Topic No.	Topic Name	Brief outline of deliverables	Book reference	Pre-requisites for preparation	Course Outcome
+	1.0	Introduction to supply chain analytics and it's application; strategic importance of it & computer simulation in supply chain management, supply chain drivers and metrics, supply chain become the value chain, responsive supply chain.	-			CO1, CO2
+	2.0	Introduction of demand management, demand planning framework and structure, time, geography, product dimension, consistency of forecasted data, supply management; inventory management; production management; transportation management; location management; sales & operations planning and execution (s&op, s&oe);	-			CO1, CO2
+	3.0	Logistics System Design, Logistics Infrastructure Development, Multiple Channel Distribution; Multi-Echelon System; Model development	-			CO3, CO4
+	5.0	Primary and Secondary Transportation; Unitising; Logistics Costing; Expressed as Ratio of Value Added or Sales; Integration of all activities for effective supply chain performance.	-			CO1, CO2, CO3, CO4
+	6.0	Logistics information System; Reverse logistics; designing of logistics network. 3PL / 4PL. Selection of logistics provider, logistics intermediaries	-			CO3, CO4

Agenda

- Introduction to supply chain analytics and it's application;
- Strategic importance of it & computer simulation in supply chain management
- Supply chain drivers and metrics
- Supply chain become the value chain
- Responsive supply chain.

Supply Chain Management



The Modern Supply Chain: From Foundational Process to Strategic Imperative

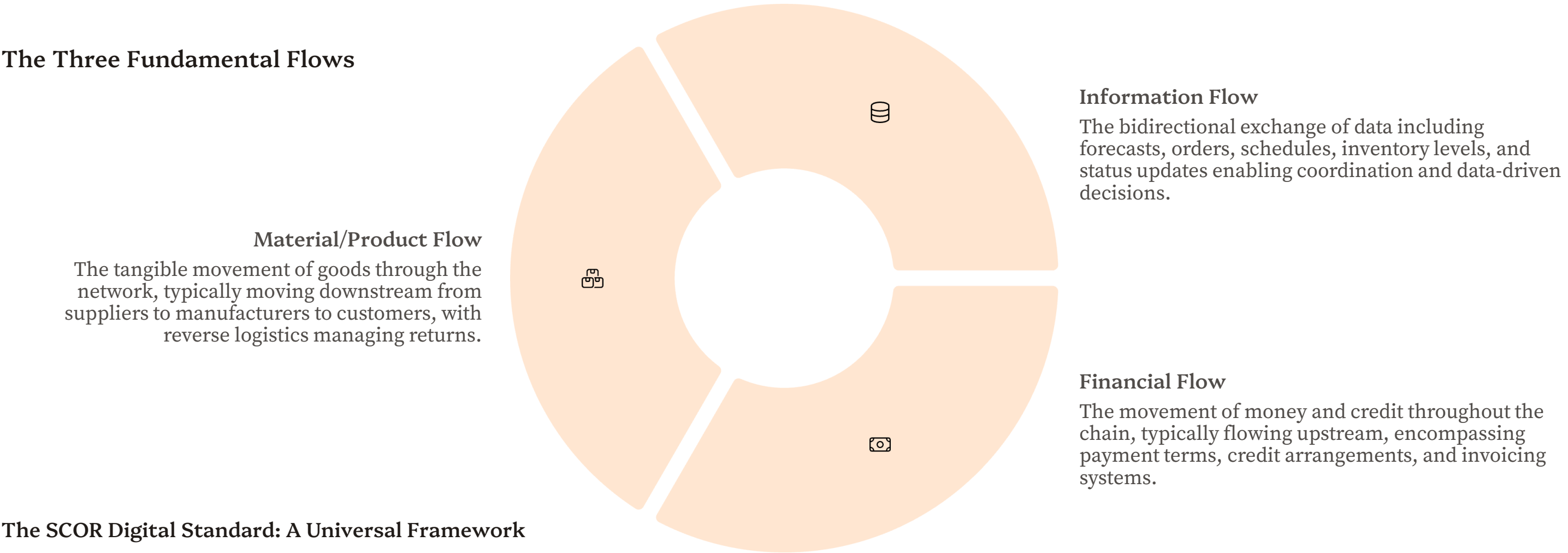
Supply Chain Management has evolved from a series of discrete operational tasks into a comprehensive and strategic management philosophy. This presentation explores how modern supply chains have become the critical engines driving commerce, ensuring products and services reach their destinations efficiently and effectively, while an increasingly complex global landscape.

Part I: The Architecture of the Modern Supply Chain

Defining the Discipline: An Integrated Management Philosophy

- A Comprehensive Strategic Discipline**
Supply Chain Management (SCM) has evolved from discrete operational tasks into a strategic discipline aimed at creating competitive advantage through the integrated management of product journeys.
- Cross-Functional Coordination**
SCM requires seamless coordination of major business functions—marketing, sales, product design, finance, and IT—within an organisation and across networks of external partners.
- Extensive Process Scope**
The scope encompasses forecasting, demand planning, procurement, supplier management, operations, manufacturing, inventory management, warehousing, and logistics.

The Three Fundamental Flows



The SCOR Digital Standard: A Universal Framework

The Supply Chain Operations Reference model provides a standardised methodology for describing, measuring, and evaluating supply chain performance. The latest SCOR Digital Standard (SCOR DS) represents a paradigm shift from a linear chain concept to a more synchronous and interconnected network model.

The Strategic Objectives: Balancing Competing Imperatives

Excellence in modern supply chain management is defined by an organisation's ability to strategically calibrate the balance between efficiency and cost reduction against resilience, agility, and customer value.

The Classic Pillars: Efficiency and Cost Reduction

Historically focused on maximising efficiency and minimising costs through lean manufacturing, Just-in-Time (JIT) inventory, and sourcing from the lowest-cost global suppliers.

Example: Walmart's business model leverages logistical techniques such as cross-docking and scale to drive down supplier costs, supporting its "Everyday Low Prices" promise.

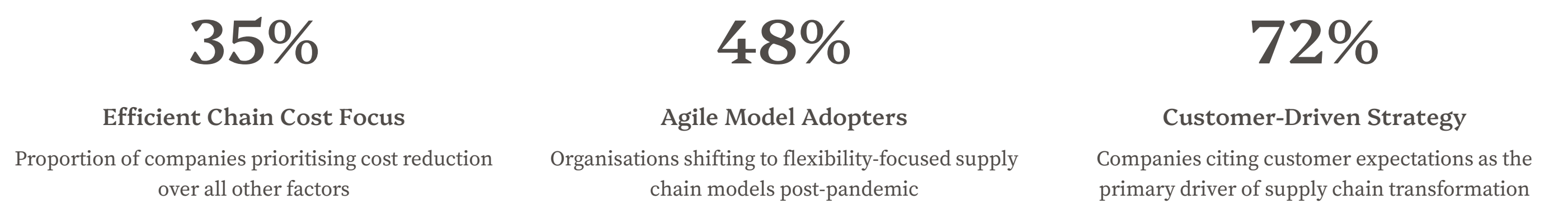
The Modern Imperatives: Resilience and Agility

Resilience: The ability to withstand, adapt to, and recover from disruptions through contingency planning and supplier diversification.

Agility: The capacity to respond quickly to unexpected changes in demand or supply, exemplified by Zara's fast-fashion model that can take designs from runway to retail stores in weeks.

The Ultimate Arbiter: Customer Satisfaction

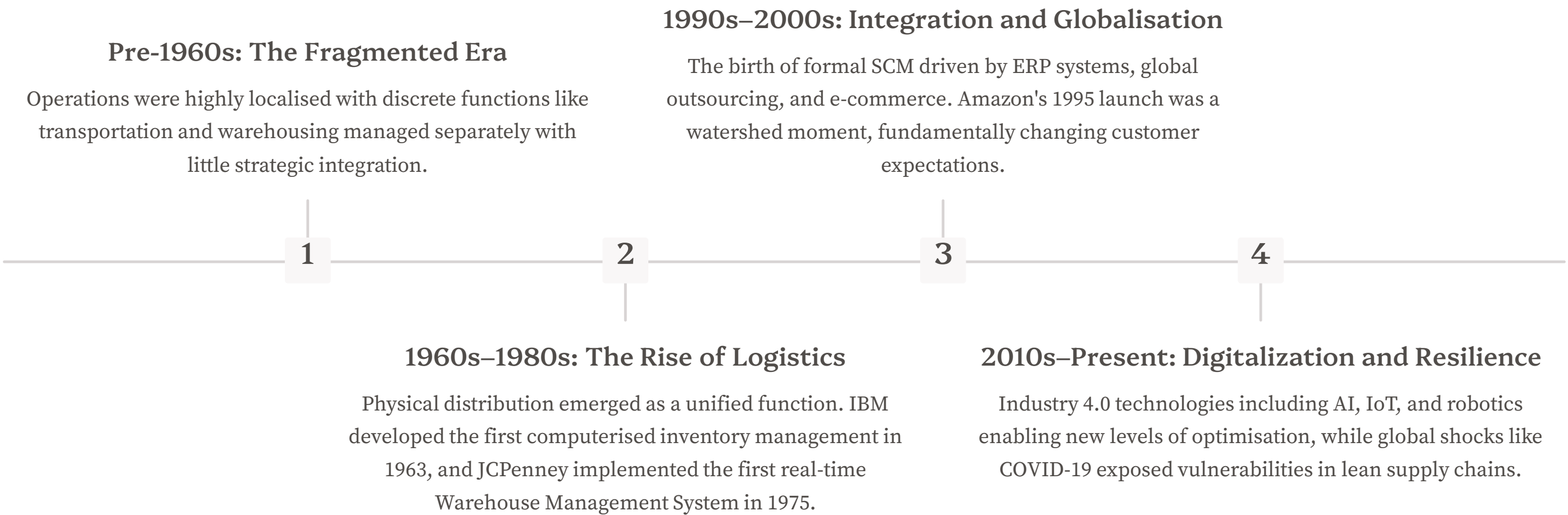
Customers now demand low prices, speed, flexibility, and transparency. Supply chains that consistently deliver on promises by ensuring product availability, quality, and timely delivery build customer loyalty and brand reputation.



The strategic tension in modern supply chain management lies in balancing competing objectives. The aggressive pursuit of maximum cost efficiency directly undermines resilience, as demonstrated during the COVID-19 pandemic when hyper-efficient supply chains were the first to break.

Part II: The Evolution and Environment of Global Supply Chains

A Strategic Evolution: From Logistics to Boardroom Imperative



Supply chain management's evolution follows a trajectory of increasing integration and abstraction, enabled by technological leaps. Each advancement expanded the scope of what could be controlled and optimised, transforming SCM from a tactical function into a source of strategic advantage.

Navigating the Modern Risk Landscape

The contemporary supply chain operates in an environment of unprecedented volatility and complexity. Risk management is no longer a secondary compliance function but a primary driver of supply chain design and strategy.



Red Sea Shipping Crisis 2024

Attacks on vessels forced rerouting around Africa, adding 10 days and 4,000 miles to Asia-Europe voyages. Freight costs surged by 350%, contributing up to 0.7 percentage points to global inflation.



Pandemic-Induced Shocks

COVID-19 created simultaneous disruption to supply and demand globally. 72% of companies reported negative effects on their supply chains, with only 2% feeling fully prepared.

Systemic Risk in Hyper-Efficient Chains

The relentless pursuit of efficiency has created complex, interconnected networks with minimal buffers. Practices like lean manufacturing, JIT inventory, and single-sourcing have systematically stripped out redundancy in the name of cost reduction.

This creates **systemic risk**: the danger that a localised disruption can trigger cascading failures across the entire network. The 2021-2022 semiconductor shortage exemplifies how disruptions at key foundries caused massive economic losses across multiple industries.

Rising Customer Expectations

The "Amazon Effect" has permanently shifted customer expectations toward unprecedented speed, transparency, and low-cost delivery. High performance is no longer a differentiator but the minimum requirement for competing, with brand loyalty becoming increasingly fragile.

This challenging environment forces a strategic re-evaluation, shifting focus from optimising stable systems to building resilient networks capable of navigating constant disruption.

The Sustainability Imperative: ESG and the Circular Economy

Supply chains are increasingly judged on their environmental, social, and governance (ESG) performance. This sustainability imperative is a core strategic issue with profound financial, operational, and reputational implications.



The Three Pillars of a Sustainable Supply Chain



The sustainability imperative is deeply intertwined with core strategic objectives. Poor ESG performance is a direct source of financial, operational, and reputational risk. A sustainable and ethical supply chain enjoys stronger brand reputation, attracts superior talent, and is more favourably viewed by investors and consumers.

Part III: Levers of Competitive Advantage

Strategic Frameworks for a Complex World

Effective supply chain management in a volatile environment requires structured, analytical approaches to decision-making. Several key frameworks have become indispensable tools for strategists.



SCOR Framework

Translates strategic goals into measurable operational metrics across five attributes: Reliability, Responsiveness, Agility, Costs, and Asset Management. Enables data-driven assessment and benchmarking against industry standards.



Total Cost of Ownership

Shifts focus from purchase price to total lifecycle costs, including acquisition, operation, and end-of-life expenses. Advanced models incorporate risk-adjusted costs for factors like geopolitical instability and IP theft exposure.

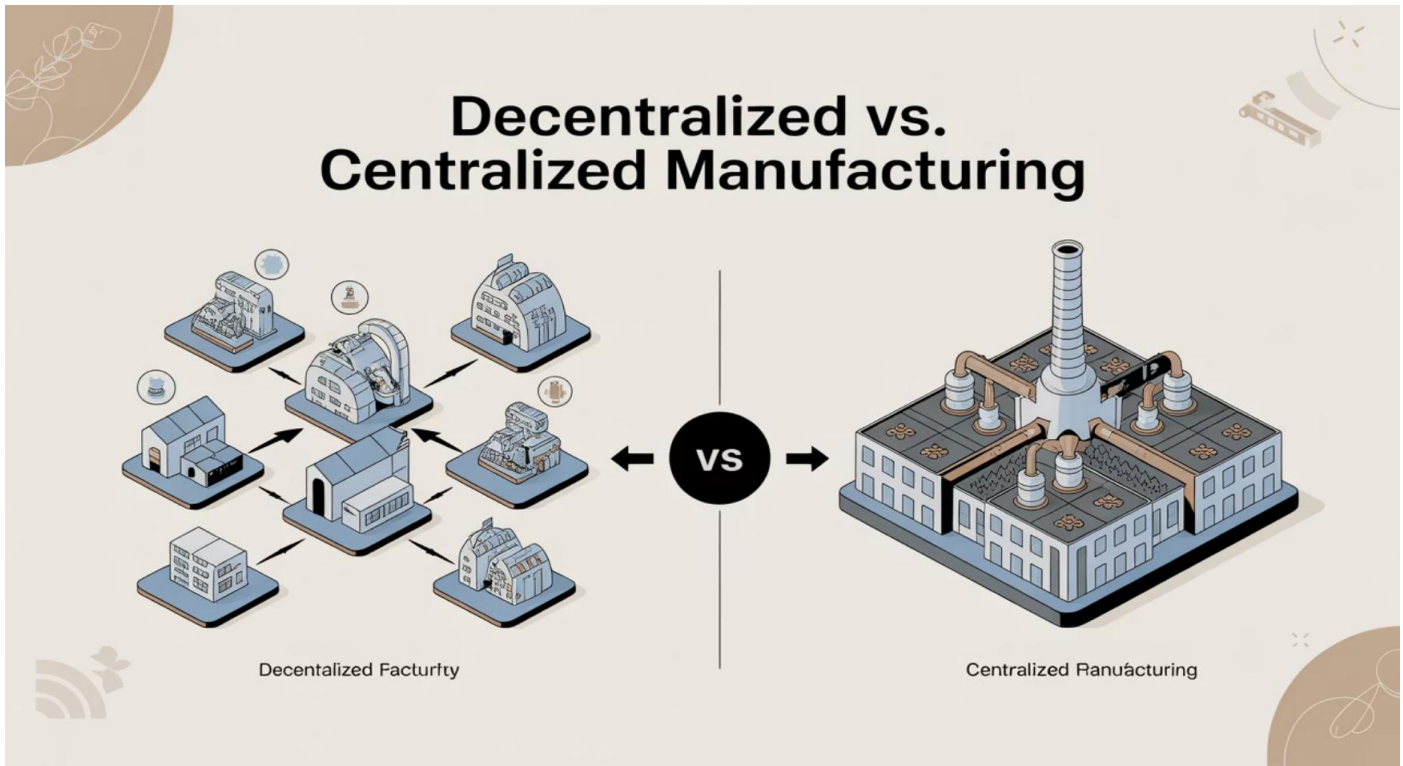


Network Design

Focuses on creating agile, resilient supply chains through decentralisation. The "China Plus One" strategy exemplifies this approach, diversifying manufacturing and sourcing beyond China to countries like Vietnam, India, and Mexico.

Agility vs Resilience: Key Distinctions

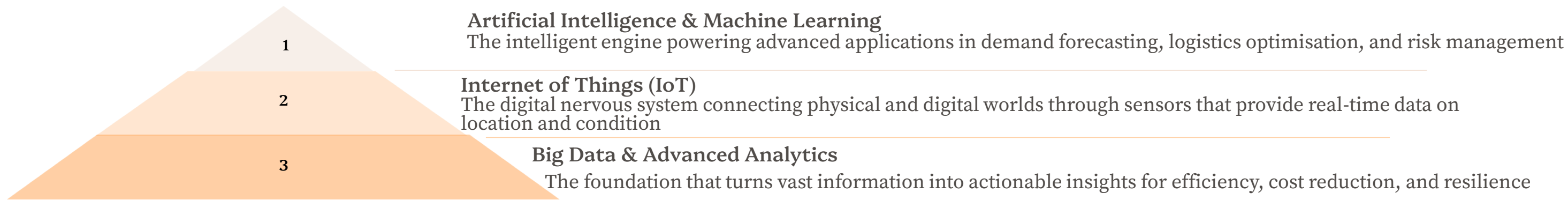
Resilience	A state of readiness achieved through proactive risk planning and diversification, enabling systems to withstand disruptions
Agility	A capability enabling rapid response to change, such as flexible manufacturing systems that can quickly switch between products



Decentralised manufacturing reduces transportation costs and lead times, improves local market responsiveness, and increases resilience by eliminating single points of failure.

The Digital Transformation of the Supply Chain

The modern supply chain is being fundamentally reshaped by digital technologies that enable a shift from reactive management to predictive and autonomous operations.



Key AI Applications in Supply Chain

- Demand Forecasting using XGBoost, Random Forest, and hybrid neural networks to analyze historical sales, market trends, and external factors
- Logistics Optimization through Reinforcement Learning for dynamic routing problems, learning optimal policies through trial and error
- Risk Management using Natural Language Processing to scan unstructured text for early warning signals of disruptions

Blockchain: The Layer of Trust

Blockchain provides a distributed, immutable ledger that enhances traceability and transparency. Walmart's implementation reduced the time to trace a package of mangoes from seven days to 2.2 seconds, enabling rapid, precise recalls during safety issues.

Digital Twin

A virtual replica of physical assets or processes continuously updated with real-time data, enabling safe "what-if" simulations without real-world risk.

Supply Chain Control Tower

A centralized hub combining people, processes, data, and technology to provide end-to-end visibility and enable collaborative, data-driven decision-making.

Integrated Systems: Digital Twins and Control Towers

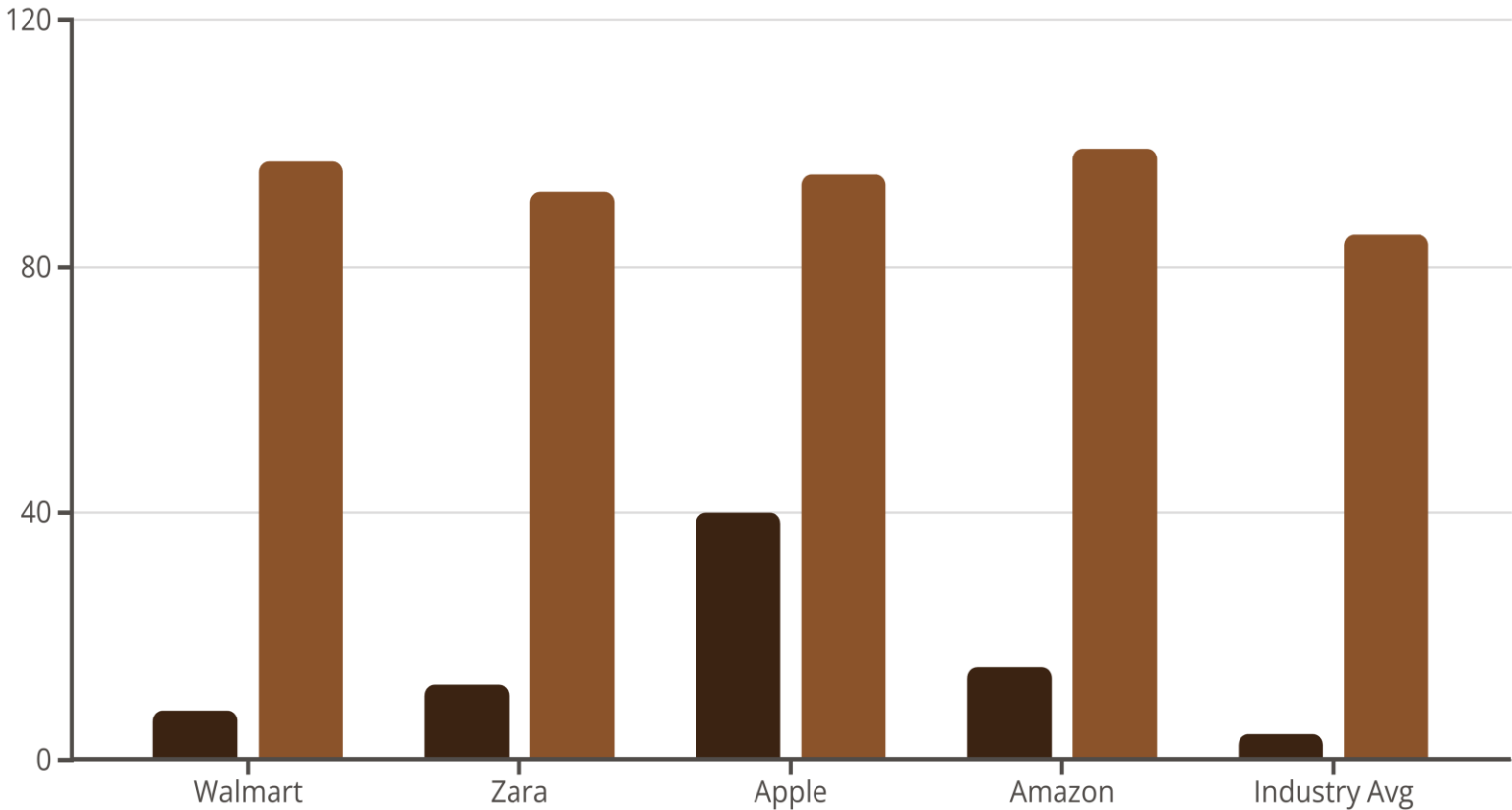


Case Studies in Strategic Execution

World-class supply chains are meticulously designed to support specific corporate strategies. Four industry titans demonstrate how different strategic objectives are realized through unique combinations of operational tactics and technology.

Walmart: Relentless Cost Efficiency
Core Strategy: Cost leadership enabling "Everyday Low Prices"
Key Tactics: Cross-docking, private truck fleet, Vendor-Managed Inventory, and continuous technology adoption
Outcome: Hyper-efficient, low-cost supply chain enabling market saturation with 90% of Americans living within 15 minutes of a store

Apple: Global Complexity Management
Core Strategy: Product innovation, premium quality, and seamless customer experience
Key Tactics: Strategic sourcing, multi-supplier resilience, vertical integration of core technology, and JIT manufacturing
Outcome: Negative cash conversion cycle (averaging -64 days) and low inventory levels (9-12 days) contributing to enormous profitability



Zara: Fast-Fashion Agility
Core Strategy: Speed and responsiveness to fast-changing fashion trends
Key Tactics: 15-day design-to-retail cycle, vertical integration, near-sourcing, and small-batch production
Outcome: 12x inventory turnover (vs. industry average of 3-4) and 85% of items sold at full price (vs. industry average of 60%)

Amazon: Technology-Driven Fulfillment
Core Strategy: Obsessive customer centricity delivered through speed
Key Tactics: Distributed fulfillment network, pioneering robotics, AI-driven logistics, and Supply Chain as a Service
Outcome: Click-to-delivery speed reduced by over 50% and increased e-commerce conversion through reliable fast-delivery estimates

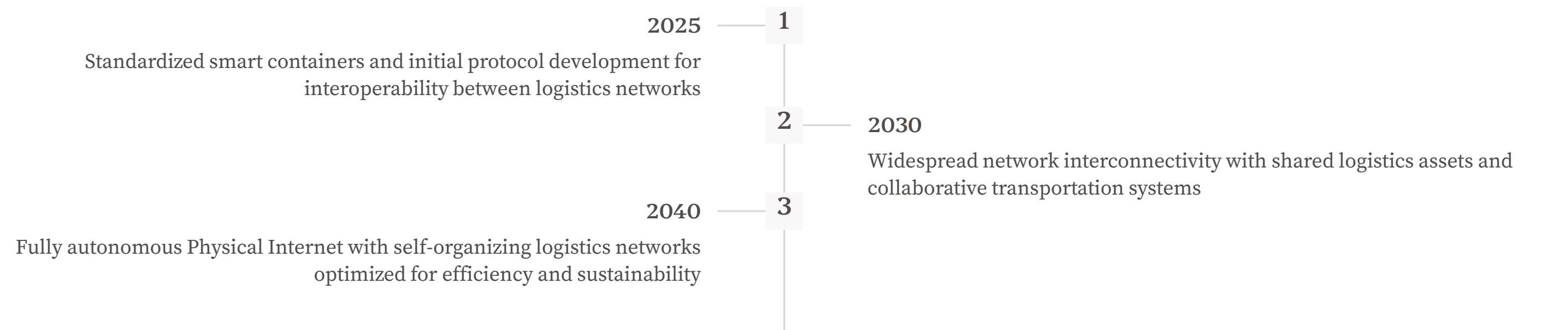
Each supply chain is uniquely configured to deliver a specific competitive advantage: Walmart on cost, Zara on speed, Apple on quality, and Amazon on customer fulfillment.

Part IV: The Future of Supply Chain Management

The Next Frontier: Emerging Concepts and Technologies

The Physical Internet (PI): Reimagining Global Logistics

The Physical Internet envisions an open, global logistics system functioning analogously to the digital internet. Physical goods in standardized, smart containers would move seamlessly across a shared network of logistics hubs, dramatically improving efficiency and sustainability.



The Rise of Autonomous Supply Chains

The convergence of AI, IoT, and robotics is paving the way for autonomous supply chains that can plan, execute, and optimize with minimal human intervention, encompassing AI-driven planning, automated procurement, "lights-out" warehouses, and autonomous delivery.

Advanced AI Applications

Beyond predictive AI, we're moving toward generative and agentic AI that can create novel solutions and take autonomous actions to achieve high-level goals, such as optimizing logistics costs while maintaining service levels.

The Indispensable Human Element

The Future of SCM Professionals

Technology isn't making human expertise obsolete but elevating it toward higher-level strategic, creative, and ethical responsibilities that are more critical than ever.

The Great Shift: From Tactical to Strategic

AI and automation are offloading repetitive, data-intensive, and transactional work, liberating human professionals to focus on high-value activities machines cannot perform well.



Strategic Decision-Making

Setting objectives and aligning supply chain strategy with broader corporate goals based on contextual understanding and long-term vision



Ethical Oversight

Designing ethical frameworks, auditing AI systems for fairness and bias, and ensuring accountability for automated decisions



Complex Problem-Solving

Navigating novel, unprecedented disruptions requiring creativity, adaptability, intuition, and empathetic leadership

New Core Competencies

- **Human-Machine Teaming:** Designing, managing, and working alongside intelligent systems, knowing when to trust AI recommendations and when to override them
- **Leadership and Emotional Intelligence:** Guiding teams through constant change and fostering a culture of continuous learning and adaptation

Emerging Supply Chain Careers



- Supply Chain Data Scientists
- AI Ethics and Bias Analysts
- Digital Twin Operators
- Robotics Fleet Managers

The paradox of the AI-driven supply chain is that it makes human judgment **more**, not less, valuable. While AI excels at "how" questions, humans must answer the far more difficult questions of "what" and "why".

The Strategic Imperative of Technology in Modern SCM

Information Technology (IT) and computer simulation have evolved from back-office support functions into core strategic pillars of effective Supply Chain Management. Together, they empower organisations to enhance efficiency, improve visibility, bolster resilience, and make smarter, data-driven decisions.

Key Benefits of IT Integration



Enhanced Visibility and Transparency

ERP, WMS, and TMS systems provide real-time data on inventory, orders, and shipments, allowing managers to track goods and proactively identify potential bottlenecks.



Improved Operational Efficiency

Automation of routine tasks reduces errors, saves time, and frees human resources for strategic activities while optimising transportation routes and inventory levels.



Superior Collaboration

Modern IT platforms break down silos and enable seamless collaboration among all supply chain partners through shared information sources.

Data-Driven Strategic Decision-Making

IT provides the foundation for advanced analytics, allowing companies to leverage Big Data, AI, and Machine Learning. These technologies analyze vast datasets to produce more accurate demand forecasts, identify emerging risks, and optimize complex processes, shifting from reactive problem-solving to proactive, predictive management.

Core IT Systems in Supply Chain Management



Enterprise Resource Planning (ERP)

The central nervous system integrating all major functions into a unified platform, providing a holistic view of the entire supply chain.



Warehouse Management Systems (WMS)

Specialised software controlling warehouse operations through barcoding and RFID for real-time tracking and process automation.



Transportation Management Systems (TMS)

Focused on logistics, optimising shipping routes, selecting cost-effective carriers, and tracking shipments in real-time.

The Power of Computer Simulation in Supply Chain Strategy

While IT provides the data and infrastructure, computer simulation provides strategic foresight. Creating a dynamic, computer-based model of a supply network allows organisations to experiment and stress-test in a risk-free environment, accounting for complex interdependencies that analytical models often cannot capture.

Key Applications and Benefits



Strategic Network Design

Modelling different scenarios for factory, warehouse, and distribution centre locations to balance cost, service levels, and lead times.



Inventory Optimization

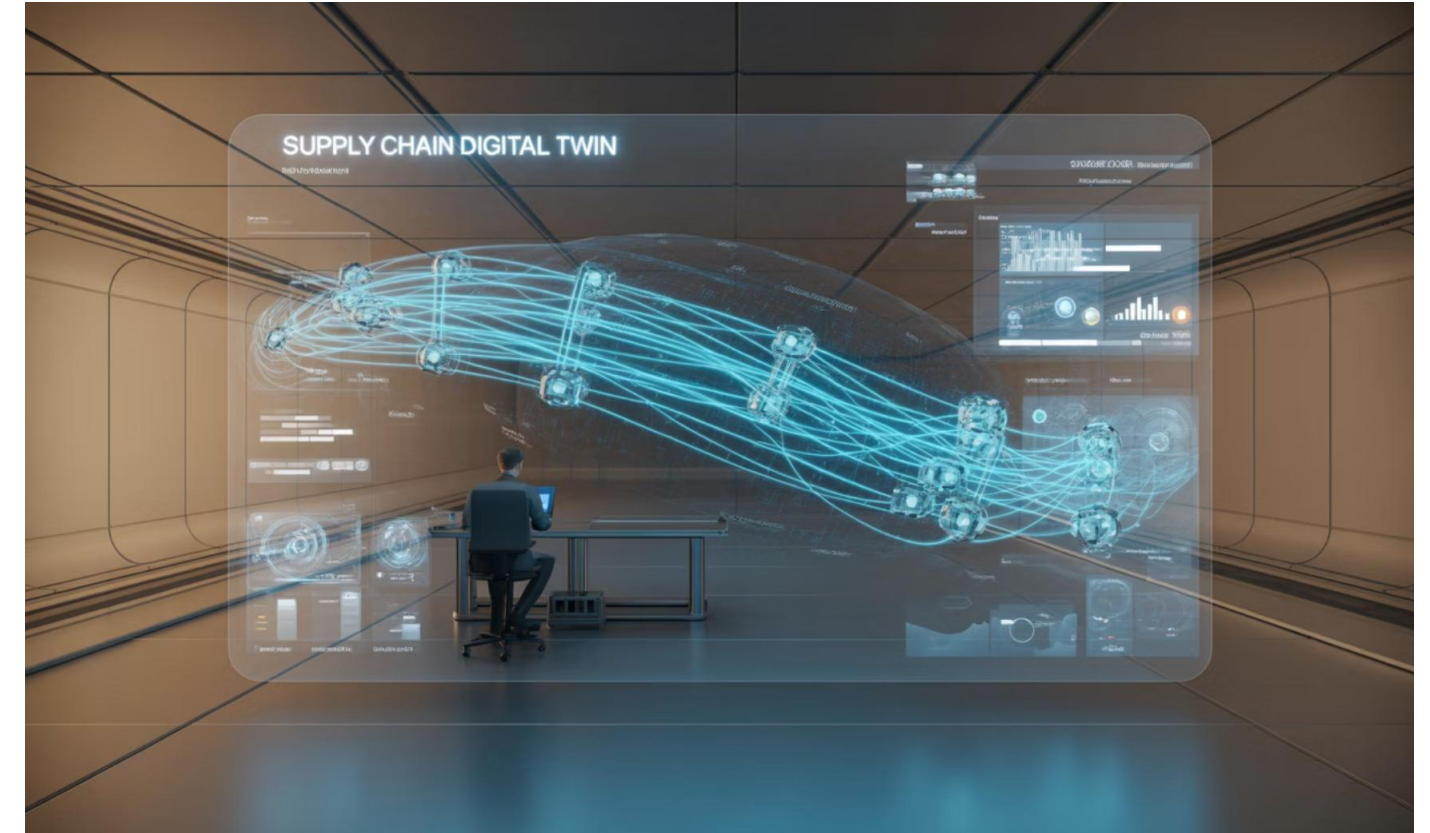
Testing various inventory policies to understand their impact on holding costs and service levels, minimising both stockouts and costly overstocks.



Risk Management

Running "what-if" scenarios to model the impact of disruptions like supplier failures or natural disasters, identifying vulnerabilities and developing contingency plans.

The Digital Twin: Ultimate Technology Synergy



A digital twin is a virtual replica of a physical supply chain continuously updated with real-time data from IT systems and IoT sensors. This creates a dynamic, high-fidelity simulation that allows managers to not only visualize current operations but also run simulations and test scenarios based on live data.

Gartner predicts that by 2028, 15% of large enterprises will have embedded intelligent simulation in their operations, with the digital twin serving as the foundation.

Implementation Challenges

Despite clear benefits, adoption faces hurdles including significant upfront costs, integration complexity with legacy systems, organisational resistance to change, and skills shortages for effectively managing these sophisticated systems.

The Six Supply Chain Drivers

Every supply chain's performance in terms of responsiveness and efficiency is determined by how well it manages **six drivers** — three logistical and three cross-functional (Chopra & Meindl framework). A firm designs the level of each driver to achieve the desired **strategic fit** between competitive strategy and supply chain strategy.

LOGISTICAL DRIVERS



Facilities

Where product is made, stored, or serviced — location, capacity and flexibility choices.



Inventory

Raw material, WIP and finished goods held across the chain to buffer against uncertainty.



Transportation

Movement of inventory between stages — mode, network and routing decisions.

CROSS-FUNCTIONAL DRIVERS



Information

Data and analysis on facilities, inventory, transport, customers used to make decisions.



Sourcing

Set of processes to procure goods/services — who performs which activity, make or buy.



Pricing

How much a firm charges for goods/services, shaping buyer behaviour and demand patterns.

DIAGRAM

Drivers of Supply Chain Performance



Reading the model

Each driver can be tuned toward **efficiency** (low cost) or **responsiveness** (speed, flexibility).

The combined setting of all six drivers determines a supply chain's overall position on the efficiency-responsiveness spectrum, and hence its strategic fit with business goals.

Each Driver Has an Efficiency–Responsiveness Trade-off

Driver	Efficiency Focus (Low Cost)	Responsiveness Focus (Speed/Flexibility)
Facilities	Few large, centralized plants/warehouses for economies of scale	Many smaller, local facilities close to customers
Inventory	Low inventory levels held across the chain	High inventory / buffer stock to absorb demand swings
Transportation	Low-cost modes: rail, sea, full-truckload (FTL)	Fast modes: air freight, less-than-truckload (LTL), premium courier
Information	Lower investment in information sharing systems	High investment in real-time data, visibility & analytics
Sourcing	Select suppliers mainly for low cost	Select suppliers mainly for speed, flexibility, reliability
Pricing	Stable, low pricing to encourage predictable demand	Dynamic pricing to manage/shape variable demand

10 Key Supply Chain Metrics — Part I

#	Metric	Formula	What It Shows
1	Inventory Turnover Ratio	$COGS \div \text{Average Inventory Value}$	How many times inventory is sold and replaced in a period; higher = more efficient inventory use.
2	Order Fill Rate	$(\text{Orders Fully Shipped} \div \text{Total Orders}) \times 100$	Percentage of customer orders fulfilled completely from available stock, first time.
3	On-Time Delivery (OTD) %	$(\text{On-Time Deliveries} \div \text{Total Deliveries}) \times 100$	Reliability of the transportation and fulfilment process against promised dates.
4	Perfect Order Rate	$\% \text{ On-time} \times \% \text{ Complete} \times \% \text{ Damage-free} \times \% \text{ Accurate Docs}$	Composite measure of flawless order execution across every dimension.
5	Cash-to-Cash Cycle Time	$DIO + DSO - DPO \text{ (in days)}$	Days between paying suppliers and collecting cash from customers; lower ties up less working capital.

10 Key Supply Chain Metrics — Part II

#	Metric	Formula	What It Shows
6	Transportation Cost % of Sales	$(Total\ Freight\ Cost \div Net\ Sales\ Revenue) \times 100$	Share of revenue consumed by logistics/transport; tracks cost efficiency of distribution.
7	Warehouse Capacity Utilization	$(Space/Volume\ Used \div Total\ Available\ Capacity) \times 100$	How intensively storage capacity is used; guides facility expansion or consolidation decisions.
8	Forecast Accuracy (via MAPE)	$MAPE = (\sum Actual - Forecast / Actual) \div n \times 100$	Error between forecasted and actual demand; lower MAPE means more reliable planning.
9	Total Supply Chain Cycle Time	<i>Sum of lead times: sourcing + production + delivery</i>	Total time from raw-material order to final customer delivery; a core speed indicator.
10	Return on Supply Chain Fixed Assets	$Supply\ Chain\ Revenue \div Net\ Value\ of\ SC\ Fixed\ Assets$	Revenue generated per rupee invested in supply chain infrastructure (plant, warehouse, fleet).

Takeaway: Metrics 1–5 emphasize execution reliability (fulfilment, delivery, cash flow), while metrics 6–10 emphasize cost, capacity and planning accuracy. Together they let managers track both the **efficiency** and **responsiveness** of the six drivers discussed earlier.

SECTION 2

From a linear cost-minimizing pipeline to an integrated, value-creating network

Why 'Supply Chain' Is Becoming 'Value Chain'

The traditional supply chain view treated logistics as a **cost centre** — a linear sequence of push activities from raw material to finished goods. Michael Porter's **Value Chain** model (1985) reframed this: every activity — inbound logistics, operations, outbound logistics, marketing & sales, and service — should add *value the customer is willing to pay for*, supported by infrastructure, HR, technology and procurement.



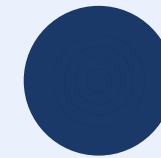
From Push to Pull

Activities are triggered by customer demand and value creation, not just by moving product forward.



Integration Over Silos

Suppliers, manufacturers, distributors and retailers act as one coordinated value-creating network.



Value, Not Just Cost

Every stage is judged by the value/margin it contributes, not only by the cost it avoids.

In short: *a supply chain moves products efficiently; a value chain uses that same movement to deliberately build competitive advantage and customer value at each step.*

DIAGRAM

Porter's Value Chain Model

SUPPORT ACTIVITIES

Firm Infrastructure (planning, finance, general management)
Human Resource Management (recruiting, training, compensation)
Technology Development (R&D, IT systems, process automation)
Procurement (purchasing inputs, supplier management)

PRIMARY ACTIVITIES

Inbound Logistics	Operations	Outbound Logistics	Marketing & Sales	Service
Receiving, warehousing & inventory control of inputs	Transforming inputs into finished goods/services	Order fulfilment, warehousing & distribution to buyers	Advertising, promotion, pricing & channel selection	Installation, repair, training & customer support

MARGIN

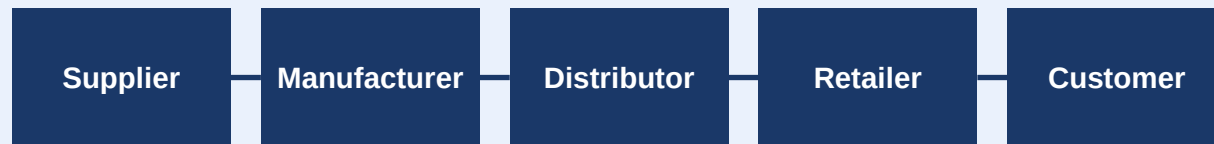
Value chain logic: Primary activities create and deliver the product; support activities enable them. The cumulative value created, minus the cost of performing these activities, determines the firm's *margin* — the essence of competitive advantage.

DIAGRAM

From a Linear Chain to a Value Network

TRADITIONAL SUPPLY CHAIN

Linear · sequential · cost-focused

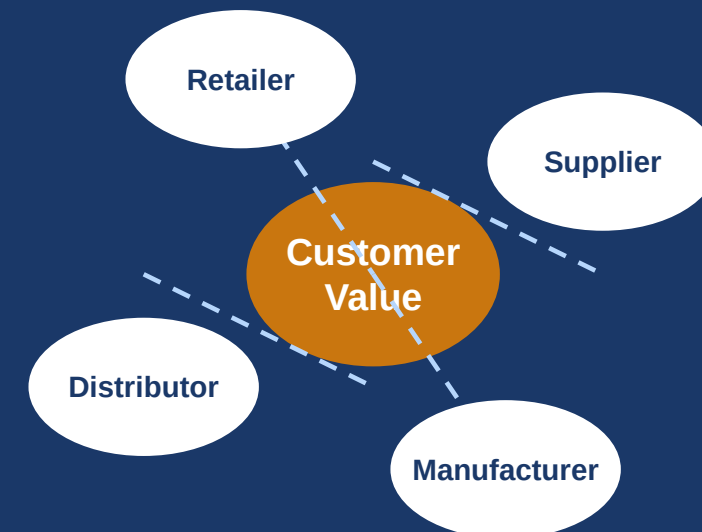


→ One-directional flow of material, driven by forecasts pushed downstream

- Goal: minimize cost
- Information flows in isolated silos
- Success = on-time, low-cost delivery

INTEGRATED VALUE CHAIN

Networked · collaborative · value-focused



- Goal: maximize value to customer
- Two-way, real-time information sharing
- Success = value created & captured jointly

SECTION 3

Designing speed and flexibility into the chain to match uncertain, fast-changing demand



Fisher's Model & the Case for Responsiveness

Marshall Fisher (1997) argued that supply chain strategy must match **product type**. **Functional products** (stable demand, long life cycles) suit **efficient** supply chains. **Innovative products** (unpredictable demand, short life cycles, high margins) need a **responsive** supply chain built for speed and flexibility rather than lowest cost.

Design characteristics of a responsive supply chain:



Speed & Agility

Short cycle times and rapid replenishment to react quickly to real demand.



Flexible Capacity

Excess/flexible capacity absorbs demand spikes instead of relying only on stock.



Postponement

Delay final assembly/customization until real demand signals arrive.



Demand-Driven (Pull)

Production and replenishment triggered by actual sales, not forecasts alone.



Buffer Where It Counts

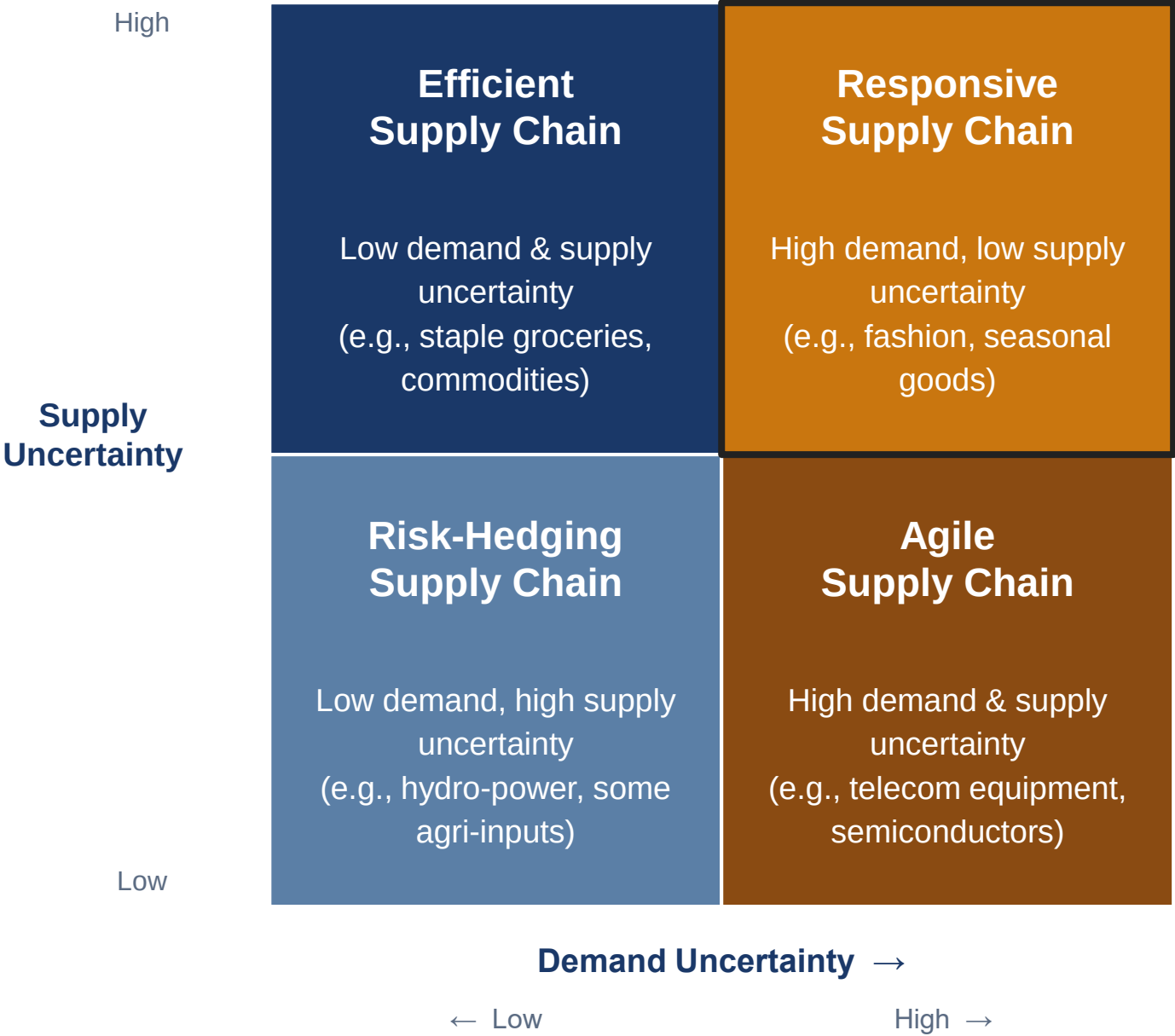
Strategic safety stock at key points to hedge against supply/demand risk.



Decentralized Decisions

Local, empowered decision-making closer to the customer for faster response.

Matching Supply Chain Strategy to Uncertainty



Focus of this session: the **Responsive Supply Chain** quadrant — used when demand is volatile/unpredictable but the supply base itself is fairly reliable, requiring speed and flexibility rather than pure cost efficiency (adapted from Fisher, 1997 and Lee, 2002).

Efficient vs Responsive Supply Chain

Dimension	Efficient Supply Chain	Responsive Supply Chain
Primary Goal	Supply demand at the lowest possible cost	Respond quickly to unpredictable demand
Manufacturing Focus	High utilization, long production runs	Flexible capacity, quick changeovers
Inventory Strategy	Minimize inventory to cut cost	Maintain buffer stock to hedge shortages
Lead Time Focus	Reduce lead time without raising cost	Aggressively reduce lead time, even at higher cost
Supplier Selection	Cost and quality first	Speed, flexibility and quality first
Product Design	Maximize performance, minimize cost	Modular design enabling postponement

KEY TAKEAWAYS



Facilities, inventory, transportation, information, sourcing and pricing shape performance — and 10 numerical metrics let managers track it objectively.



Supply chains are evolving into value chains: integrated, customer-focused networks judged by value created, not just cost avoided.



For uncertain, innovative-product demand, a responsive supply chain — fast, flexible, postponed — outperforms a purely cost-efficient one.

Thank You